🏛 GLA UNIVERSITY, MATHURA — DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

HPC Masterclass | Day 5 Post-Assessment Final Examination

Course: B.Tech CSE 7th Sem — High Performance Computing Masterclass
Module: Day 5 — Generative AI, LLMs, Intel Gaudi & OpenVINO Inference Optimization
Duration: 60 Minutes | **Total Questions**: 50 MCQs | **Maximum Marks**: 50

Student Name: _____ **Roll Number**: _____ **Section**: _____

📝 Instructions for Candidates

1. All questions are compulsory. Each question carries exactly 1 mark.
2. Select the single best answer for each question.
3. No negative marking.

Q01. A dual-socket 4th Gen Intel Xeon Scalable 8480+ server has 112 physical cores and 8 DDR5-4800 memory channels delivering 307.2 GB/s theoretical peak bandwidth. When running autoregressive token generation on a 7-billion parameter LLM with INT4 weight quantization (model weight size = 3.5 GB), assuming an 80% memory bandwidth efficiency factor and batch size = 1, what is the maximum theoretical generation throughput in Tokens Per Second (TPS)?

- ▢ **(a)** 12.5 TPS
- ▢ **(b)** 70.2 TPS (calculated as (307.2 * 0.80) / 3.5 GB)
- ▢ **(c)** 350.0 TPS
- ▢ **(d)** 1,024.0 TPS

Q02. What is the physical architectural structure of the Intel Advanced Matrix Extensions (AMX) TMUL engine in 4th/5th Gen Intel Xeon processors?

- ▢ **(a)** A 1D vector ring of ALU pipelines operating on 64-bit scalar registers.
- ▢ **(b)** Eight 2-dimensional physical tile registers (TMM0 to TMM7), each capable of holding up to 1,024 bytes (16 rows by 64 bytes), paired with a systolic grid of multiply-accumulate units executing TDPBF16PS or TDPBUSD instructions.
- ▢ **(c)** A dedicated optical co-processor connected over PCIe Gen 3.
- ▢ **(d)** An FPGA bitstream loaded into the L3 cache slice.

Q03. In the Habana Gaudi 2 architecture, what is the exact aggregate bi-directional scale-out network bandwidth provided per accelerator chip using its integrated RoCE v2 ports?

- ▢ **(a)** 100 Gbps total
- ▢ **(b)** 2.4 Tbps (24 ports * 100 GbE RoCE v2 integrated on-die)

- ▢ **(c)** 10 Gbps shared across the PCIe bus
- ▢ **(d)** 100 MB/s over USB 3.0

Q04. How does the Habana Gaudi 3 architecture improve compute and memory specifications over Gaudi 2?

- ▢ **(a)** It switches from silicon to vacuum tubes.
- ▢ **(b)** It doubles MME matrix compute capacity (4 to 8 MMEs), expands TPCs from 8 to 64, increases HBM2e/HBM3 memory to 128 GB at 3.7 TB/s bandwidth, and upgrades the 24 on-die networking ports to 200 GbE RoCE.
- ▢ **(c)** It removes all on-die networking to save manufacturing costs.
- ▢ **(d)** It replaces HBM with standard DDR4 SODIMMs.

Q05. In OpenVINO Intermediate Representation (IR), what fundamental graph optimization is performed when an `Add` (bias) operation immediately follows a `MatMul` operation?

- ▢ **(a)** The model weights are rounded to zero.
- ▢ **(b)** Operator Fusion: The OpenVINO Model Optimizer fuses the MatMul and Add nodes into a single composite `FullyConnected` primitive with a fused bias post-operation, eliminating an intermediate memory write and read cycle.
- ▢ **(c)** The two layers are executed on separate CPU cores concurrently.
- ▢ **(d)** The computational graph is converted into an audio waveform.

Q06. When quantizing a pre-trained transformer model using Intel NNCF with Weight-Only INT4 Quantization, what is the role of the calibration dataset?

- ▢ **(a)** It trains the model for 100 additional epochs using AdamW.
- ▢ **(b)** It passes representative domain token sequences through the model to record activation distributions across layers, allowing advanced algorithms (like AWQ or SmoothQuant) to detect activation outliers and determine optimal weight quantization scales and zero-points.
- ▢ **(c)** It checks student grades against the university database.
- ▢ **(d)** It deletes all layers with negative weights.

Q07. Consider a linear layer with input dimension $d_{in} = 4,096$ and output dimension $d_{out} = 4,096$. In full fine-tuning, 16,777,216 parameters are updated. If LoRA is applied with rank $r = 16$, how many trainable parameters exist in this layer's adapter matrices A and B?

- ▢ **(a)** 131,072 parameters (a 99.22% parameter reduction)
- ▢ **(b)** 1,048,576 parameters
- ▢ **(c)** 4,096 parameters
- ▢ **(d)** 8,388,608 parameters

Q08. In a production Slurm cluster with 4 compute nodes, each containing 8 Habana Gaudi 2 accelerators (32 HPUs total), what is the optimal Slurm batch script directive configuration to launch a distributed PyTorch training job with 1 task per HPU?

- ▢ **(a)** `#SBATCH --nodes=1 --ntasks=1 --gres=gpu:1`
- ▢ **(b)** `#SBATCH --nodes=4 --ntasks-per-node=8 --gres=habana:8`
- ▢ **(c)** `#SBATCH --nodes=32 --ntasks=1 --partition=debug`
- ▢ **(d)** `#SBATCH --cpus-per-task=1024 --mem=1TB`

Q09. What is the primary operational purpose of the Intel Extension for PyTorch (IPEX) in deep learning workflows?

- ▢ **(a)** To replace PyTorch with Microsoft Excel macros.
- ▢ **(b)** To provide drop-in hardware acceleration for PyTorch on Intel architectures, exposing automated mixed precision (BF16/INT8), graph fusions, weight prepacking, and native Intel AMX TMUL kernel execution.
- ▢ **(c)** To convert Python into JavaScript for web deployment.
- ▢ **(d)** To manage printer queues in the server room.

Q10. Why is the prompt processing phase ('prefill') in LLMs capable of achieving near-peak TFLOPs on Intel AMX or Gaudi MMEs, whereas the single-token autoregressive decoding phase achieves < 5% of peak TFLOPs?

- ▢ **(a)** Because decoding uses a secret unoptimized algorithm.
- ▢ **(b)** Because prefill stacks all prompt tokens into a single large GEMM where arithmetic intensity is high (tokens re-use weights in cache/registers), whereas single-token decode is a matrix-vector GEMV where every single weight parameter must be loaded from DRAM to compute just 2 FLOPs per parameter.
- ▢ **(c)** Because AMX tiles automatically power off during decode.
- ▢ **(d)** Because prefill runs in C++, while decode runs in Python.

Q11. In the OpenVINO GenAI C++/Python API (`openvino_genai.LLMPipeline`), how does the pipeline manage KV-cache memory internally during multi-turn conversations?

- ▢ **(a)** It writes all KV-tensors to a temporary `.txt` file on disk.
- ▢ **(b)** It manages stateful KV-cache tensors directly inside the OpenVINO compiled model runtime memory buffers, dynamically appending new token states and evicting expired states without expensive Python-to-C++ tensor marshal copies.
- ▢ **(c)** It deletes the KV-cache after every emitted character.
- ▢ **(d)** It requires the client browser to store the KV-cache in local cookies.

Q12. When deploying an LLM with INT4 Weight-Only quantization, why is Symmetric Quantization computationally faster during runtime dequantization compared to Asymmetric Quantization?

- ▢ **(a)** Symmetric quantization requires no memory storage at all.
- ▢ **(b)** In symmetric quantization, the zero-point $Z = 0$, meaning dequantization requires only a single multiplication: $x \approx q \times S$. Asymmetric quantization requires an extra subtraction: $x \approx (q - Z) \times S$, which adds instruction overhead and register pressure inside the tight inner loop.
- ▢ **(c)** Symmetric quantization can only be run on quantum processors.
- ▢ **(d)** Asymmetric quantization causes CPU cache lines to invalidate.

Q13. What is the primary role of the `TDPBF16PS` instruction in the x86-64 instruction set on Intel Sapphire Rapids processors?

- ▢ **(a)** To compute floating-point square roots.
- ▢ **(b)** To perform a systolic dot-product of two pairs of Bfloat16 numbers from two 2D tile registers and accumulate the single-precision FP32 result into a destination 2D tile register.
- ▢ **(c)** To transfer data over the PCI Express bus.
- ▢ **(d)** To flush the CPU instruction pipeline.

Q14. In distributed LLM fine-tuning using DeepSpeed ZeRO-2 on a multi-node cluster, which components of the training state are partitioned across data-parallel ranks?

- ▢ **(a)** Only the student code comments.
- ▢ **(b)** Both the Optimizer States (Adam momentum and variance) AND the Gradients are partitioned across all ranks, while model parameters are replicated.
- ▢ **(c)** The entire model parameters, gradients, and optimizer states are partitioned.
- ▢ **(d)** No states are partitioned; everything is duplicated.

Q15. What is the key advantage of Habana's integrated on-die RoCE networking over external InfiniBand PCIe adapters during distributed All-Reduce gradient synchronization?

- ▢ **(a)** It uses cheaper copper HDMI cables.
- ▢ **(b)** It completely bypasses the PCIe bus, host root complex, and operating system kernel, streaming tensors directly from HBM memory to remote accelerator HBM memory via on-die RDMA network engines with sub-microsecond latency.
- ▢ **(c)** It eliminates the need for electric power in the server rack.
- ▢ **(d)** It automatically compresses training datasets.

Q16. In continuous batching LLM serving architectures, why is Static Padding eliminated?

- ▢ **(a)** Because padding tokens are forbidden by the HTTP/2 specification.
- ▢ **(b)** Because sequences are not combined into a single static 2D rectangular tensor; instead, requests are managed independently in dynamic token pools (e.g. via PagedAttention), evaluating only valid tokens at each decoding iteration.
- ▢ **(c)** Because transformers cannot compute attention on padding tokens.
- ▢ **(d)** Because tokenizers no longer use vocabulary IDs.

Q17. In the context of Intel NNCF, what is the effect of setting `group_size = -1` versus `group_size = 128` during INT4 Weight-Only quantization?

- ▢ **(a)** `group_size = -1` enables per-channel quantization (one scale and zero-point for the entire channel), whereas `group_size = 128` partitions the channel into blocks of 128 elements, each with its own scale factor, significantly improving accuracy at a slight metadata memory cost.
- ▢ **(b)** `group_size = -1` deletes all weights with negative values.
- ▢ **(c)** `group_size = 128` crashes the OpenVINO runtime.
- ▢ **(d)** `group_size = -1` uses 1 bit per weight.

Q18. In a FastAPI microservice serving an OpenVINO LLM, what is the purpose of running the model inference loop inside a dedicated background worker thread or process using `asyncio.to_thread()`?

- ▢ **(a)** To bypass user authentication checks.
- ▢ **(b)** Because OpenVINO model inference is CPU-bound and synchronous; running it directly on the main asynchronous event loop would block the entire event loop, freezing all concurrent HTTP connections and health check probes.
- ▢ **(c)** To convert the server into a database.
- ▢ **(d)** To force the client to download the model `.bin` file.

Q19. What is the role of the `alpha` hyperparameter in LoRA fine-tuning ($\Delta W = \frac{\alpha}{r} B \cdot A$)?

- ☐ **(a)** It sets the transparency level of the user interface.
- ☐ **(b)** It is a constant scaling factor that stabilizes training dynamics when experimenting with different values of rank r, ensuring that the magnitude of the adapter update ΔW does not drastically fluctuate when the rank is changed.
- ☐ **(c)** It sets the number of CPU threads allocated to data loading.
- ☐ **(d)** It determines the learning rate decay schedule.

Q20. Why does a 13-billion parameter FP16 model require at least 26 GB of memory just to load its weights, and how does NNCF INT4 quantization alter this constraint?

- ☐ **(a)** FP16 requires 2 bytes/param ($13 \times 10^9 \times 2 = 26$ GB); INT4 requires 0.5 bytes/param ($13 \times 10^9 \times 0.5 = 6.5$ GB), allowing a 13B model to fit comfortably into the memory of a single client workstation or edge server.
- ☐ **(b)** FP16 requires 16 bytes per parameter, whereas INT4 requires 4 bytes per parameter.
- ☐ **(c)** INT4 requires 52 GB of memory.
- ☐ **(d)** Quantization has no effect on memory footprint.

Q21. In Habana SynapseAI, what does the environment variable `PT_HPU_LAZY_MODE=0` do when debugging PyTorch code?

- ☐ **(a)** It speeds up training by 100x.
- ☐ **(b)** It disables the lazy graph compiler and forces PyTorch to execute operations eagerly on the HPU, allowing developers to set standard Python breakpoints and inspect intermediate tensor values at the cost of lower throughput.
- ☐ **(c)** It powers off all server cooling fans.
- ☐ **(d)** It converts the code to run on an Intel Core i3 laptop.

Q22. What is the primary difference between Hugging Face `PeftModel` and merging LoRA weights back into the base model using `model.merge_and_unload()`?

- ☐ **(a)** `PeftModel` can only run on Android; merged models can only run on iOS.
- ☐ **(b)** `PeftModel` maintains base weights W_0 and adapter matrices A, B separately at runtime, computing two parallel forward paths; `merge_and_unload()` explicitly adds $\frac{\alpha}{r}(B \cdot A)$ directly into W_0 in memory, producing a single standalone model with zero inference latency overhead.
- ☐ **(c)** Merging weights deletes the attention mechanism.
- ☐ **(d)** Merged models require twice as much memory during inference.

Q23. How does Intel AMX accelerate INT8 quantization workloads using the `TDPBUSD` instruction?

- ☐ **(a)** By downloading pre-compiled weights from the internet.
- ☐ **(b)** It performs dot products of signed bytes from one tile register with unsigned bytes from a second tile register, accumulating the intermediate products into 32-bit signed integer accumulators without numerical overflow.
- ☐ **(c)** By dividing two floating-point numbers.
- ☐ **(d)** By converting text tokens into Unicode binary.

Q24. In Slurm job submission, what is the role of the environment variable `SLURM_PROCID`?

- ☐ **(a)** It specifies the university course code.

- ▢ **(b)** It provides the global rank (process ID from 0 to N-1) of the current task within the active Slurm job step, used to initialize PyTorch distributed process groups (`rank=int(os.environ['SLURM_PROCID'])`).
- ▢ **(c)** It records the room temperature of the server room.
- ▢ **(d)** It sets the password for SSH login.

Q25. What is the technical cause of 'Perplexity Degradation' when quantizing an LLM to INT4 using naive Round-To-Nearest (RTN) without grouping or outlier protection?

- ▢ **(a)** The CPU runs out of disk storage space.
- ▢ **(b)** Outlier weights and activations with large magnitudes force the uniform quantization scale factor to be excessively large, collapsing the vast majority of smaller weights into 0 or 1 quantization bin and destroying the subtle semantic representations of the network.
- ▢ **(c)** The Linux operating system kernel drops network packets.
- ▢ **(d)** The model produces outputs only in uppercase letters.

Q26. In OpenVINO runtime, what is the primary benefit of enabling dynamic shapes (`ov::Dimension()`) for LLM inference?

- ▢ **(a)** It forces the model to use 100% of the GPU at all times.
- ▢ **(b)** It allows the compiled model to accept input prompts and context sequences of arbitrary variable lengths without requiring recompilation or padding to a fixed maximum sequence length.
- ▢ **(c)** It turns the neural network into an image classifier.
- ▢ **(d)** It disables the transformer feedforward layers.

Q27. In Habana Gaudi's SynapseAI stack, what is the 'TPC LLVM Compiler'?

- ▢ **(a)** A web browser designed for supercomputers.
- ▢ **(b)** A specialized backend compiler based on LLVM that compiles high-level C/C++ tensor kernels into VLIW SIMD machine code executed natively by the Tensor Processor Cores.
- ▢ **(c)** A database query optimizer.
- ▢ **(d)** A utility for formatting Linux ext4 partitions.

Q28. What is the primary difference between GPTQ (Generalized Post-Training Quantization) and AWQ (Activation-aware Weight Quantization)?

- ▢ **(a)** GPTQ uses second-order Taylor expansion (Hessian matrix inverse) to greedily quantize weights while updating remaining unquantized weights to compensate for quantization error; AWQ does not alter weight values, but identifies salient weight channels based on activation magnitudes and protects them with per-channel scaling.
- ▢ **(b)** GPTQ only works on image models; AWQ only works on audio.
- ▢ **(c)** AWQ requires retraining with SGD for 50 epochs; GPTQ does not.
- ▢ **(d)** GPTQ produces 32-bit floats; AWQ produces 64-bit doubles.

Q29. In an LLM benchmark report, what does the metric 'P99 Inter-Token Latency' represent?

- ▢ **(a)** The average token generation latency across all users.
- ▢ **(b)** The 99th percentile threshold of latency between consecutive generated tokens; 99% of all emitted tokens experience an ITL equal to or faster than this value, revealing worst-case jitter and user-perceived stutter.

- ▢ **(c)** The percentage of tokens that contained spelling mistakes.
- ▢ **(d)** The total power consumption of the server rack.

Q30. What is 'FlashAttention' and what core hardware architectural mechanism does it exploit?

- ▢ **(a)** An Adobe Flash plugin for web browsers.
- ▢ **(b)** An exact attention algorithm that tiles the computation of the softmax reduction across fast on-chip SRAM cache (L1/Shared Memory) to avoid materializing the full $N \times N$ intermediate attention matrix in slow HBM/DRAM, reducing memory I/O from quadratic to linear.
- ▢ **(c)** A method to flash new firmware onto hard drives.
- ▢ **(d)** A quantization scheme that rounds all weights to 0 or 1.

Q31. In a dual-socket Intel Xeon system with 64 cores per socket (128 total cores), how does configuring OpenVINO with two CPU streams (`ov::num_streams = 2`) mapped to NUMA nodes improve aggregate serving throughput?

- ▢ **(a)** It disables one of the CPU sockets to save power.
- ▢ **(b)** It pins one inference execution stream exclusively to Socket 0 and its local DDR5 memory, and the second stream to Socket 1 and its local DDR5 memory, completely eliminating cross-socket UPI interconnect congestion.
- ▢ **(c)** It forces both sockets to execute the exact same token in lockstep.
- ▢ **(d)** It converts the CPU into an Ethernet router.

Q32. In distributed LLM training, what is the role of 'Pipeline Parallelism' (PP)?

- ▢ **(a)** Running water pipes through the server rack for liquid cooling.
- ▢ **(b)** Partitioning the sequential layers of a large neural network across multiple accelerators (e.g. layers 1-8 on Device 0, layers 9-16 on Device 1), passing intermediate activation tensors downstream during the forward pass and gradients upstream during the backward pass.
- ▢ **(c)** Executing Python scripts line by line.
- ▢ **(d)** Compressing training data before storing it on disk.

Q33. In the Slurm workload manager, what does the command `sbatch run_gaudi.slurm` return upon successful execution?

- ▢ **(a)** The entire training dataset in CSV format.
- ▢ **(b)** A confirmation message containing the unique assigned Slurm Job ID (e.g., `Submitted batch job 1048576`).
- ▢ **(c)** The root password for the compute nodes.
- ▢ **(d)** A compiled C++ binary.

Q34. What is the primary motivation for using Asymmetric INT4 quantization on the output projections of LLM MLP feed-forward layers (e.g., `down_proj` or `fc2`)?

- ▢ **(a)** MLP layers do not contain weights.
- ▢ **(b)** The inputs to these layers are post-activation features (e.g., SwiGLU or GELU outputs) that exhibit strictly non-negative or highly skewed value distributions where asymmetric zero-point offsets prevent precision loss.
- ▢ **(c)** Asymmetric quantization uses less disk space than symmetric quantization.
- ▢ **(d)** It allows the server to run without RAM.

Q35. What is the purpose of the `hl-smi -q` command in a Habana Gaudi server environment?

- ▢ **(a)** To format all hard drives.
- ▢ **(b)** To query and display comprehensive XML/text telemetry including HPU board serial numbers, driver versions, PCIe bus IDs, memory usage, power consumption, clock frequencies, and thermal sensor readings.
- ▢ **(c)** To compile a PyTorch script into machine code.
- ▢ **(d)** To shut down the cluster.

Q36. In a distributed deep learning cluster, what is the primary symptom of 'Straggler Nodes' during synchronous All-Reduce training?

- ▢ **(a)** The server room temperature drops to zero degrees.
- ▢ **(b)** All healthy high-speed nodes complete their forward/backward compute steps quickly and sit idle waiting at the synchronization barrier for the single slowest node, causing overall cluster throughput to degrade to the speed of the straggler.
- ▢ **(c)** The model produces perfect validation accuracy on epoch 1.
- ▢ **(d)** The Slurm queue automatically cancels all user jobs.

Q37. What is the primary role of the `AutoTokenizer` class from Hugging Face when deploying an LLM with OpenVINO?

- ▢ **(a)** To generate artificial images from text descriptions.
- ▢ **(b)** To load the pre-trained token vocabulary, special token mappings (BOS, EOS, PAD), and subword merge rules, transforming raw input prompt strings into input ID tensors and attention masks expected by the OpenVINO model.
- ▢ **(c)** To convert Python into C++ code.
- ▢ **(d)** To monitor network bandwidth usage.

Q38. In the context of the Roofline Model, what does the 'Ridge Point' represent?

- ▢ **(a)** The physical height of the server rack in centimeters.
- ▢ **(b)** The minimum arithmetic intensity (FLOPs/byte) at which an application can achieve the theoretical peak floating-point compute performance of the processor, calculated as $\frac{\text{Peak Compute Performance (GFLOPs/s)}}{\text{Peak Memory Bandwidth (GB/s)}}$.
- ▢ **(c)** The maximum temperature before the CPU shuts down.
- ▢ **(d)** The total number of memory channels on the motherboard.

Q39. How does Intel NNCF implement 'SmoothQuant' to enable accurate 8-bit weight and activation quantization in LLMs?

- ▢ **(a)** By rounding all weights to even numbers.
- ▢ **(b)** By applying a mathematically equivalent per-channel scaling transformation that migrates quantization difficulty from activations to weights ($W' = W \cdot \text{diag}(s)$, $X' = X \cdot \text{diag}(s)^{-1}$), smoothing activation outliers so both weights and activations can be quantized to INT8.
- ▢ **(c)** By increasing the learning rate by 10x.
- ▢ **(d)** By converting all matrices into diagonal matrices.

Q40. In Habana Gaudi systems, what does the term 'HBM2e' stand for, and why is it located on the same interposer package as the compute die?

- ▢ **(a)** Hard Block Memory 2 Enterprise; to save disk space.
- ▢ **(b)** High Bandwidth Memory 2 Extended; mounted via a silicon interposer with a wide 1024-bit memory bus to provide multi-terabyte-per-second memory bandwidth with ultra-low trace capacitance and power consumption.
- ▢ **(c)** Hybrid Binary Matrix; to emulate quantum bits.
- ▢ **(d)** High Battery Module; to provide backup power during outages.

Q41. What is the primary function of the OpenVINO Model Server (OVMS) in an enterprise AI deployment?

- ▢ **(a)** To provide a high-throughput, C++ based microservice supporting gRPC and REST APIs (compatible with KServe and Triton protocols) for hosting optimized OpenVINO and vLLM models in production Kubernetes environments.
- ▢ **(b)** To compile Python scripts into executable `.exe` files for Windows.
- ▢ **(c)** To back up server databases to tape drives.
- ▢ **(d)** To design user interface websites.

Q42. In a distributed training job using PyTorch FSDP (Fully Sharded Data Parallel), what is the purpose of 'Activation Checkpointing' (also known as Gradient Checkpointing)?

- ▢ **(a)** Saving student test scores to disk every 5 minutes.
- ▢ **(b)** Discarding intermediate layer activations during the forward pass and recomputing them on-the-fly during the backward pass, trading ~20-30% extra compute time for a dramatic (up to 70%) reduction in activation memory.
- ▢ **(c)** Backing up the Linux kernel configuration to GitHub.
- ▢ **(d)** Writing logs to the console.

Q43. What is the primary microarchitectural advantage of Bfloat16 over FP32 when executing GEMM operations on Intel AMX TMUL units?

- ▢ **(a)** BF16 units deliver 2x higher arithmetic throughput per clock cycle compared to FP32, while consuming half the register and memory bandwidth, without sacrificing the FP32 dynamic exponent range.
- ▢ **(b)** BF16 uses zero electrical power.
- ▢ **(c)** BF16 can represent 100 times larger numbers than FP32.
- ▢ **(d)** BF16 does not require memory controllers.

Q44. In OpenVINO INT4 model serving, why is the memory footprint of the KV-cache often larger than the model weights themselves when serving large concurrency (e.g. batch size = 64, context length = 4,096 tokens)?

- ▢ **(a)** Because the KV-cache is intentionally padded with random numbers.
- ▢ **(b)** Because the weights are statically compressed to INT4 (e.g. 3.5 GB for a 7B model), whereas the KV-cache grows linearly with batch size and context length in FP16 ($64 \times 4096 \times 32\ \text{layers} \times 4096\ \text{dim} \times 2\ \text{bytes} \approx 68.7$ GB), dwarfing the compressed weight footprint.
- ▢ **(c)** Because OpenVINO duplicates the KV-cache across all hard drives.
- ▢ **(d)** Because KV-cache cannot be stored in RAM.

Q45. Which of the following Slurm environment variables indicates the total number of compute nodes allocated to the active job?

- ▢ **(a)** `SLURM_JOB_NUM_NODES`
- ▢ **(b)** `SLURM_TOTAL_DISKS`
- ▢ **(c)** `SLURM_RAM_PER_SOCKET`
- ▢ **(d)** `SLURM_USER_TIER`

Q46. In LLM generation sampling, what is the role of the 'Temperature' hyperparameter?

- ▢ **(a)** It regulates the physical cooling fans inside the server chassis.
- ▢ **(b)** It scales the raw logit values before applying Softmax ($p_i = \frac{e^{z_i/T}}{\sum e^{z_j/T}}$), where $T < 1.0$ sharpens probabilities toward greedy high-confidence tokens, and $T > 1.0$ flattens probabilities toward higher diversity and randomness.
- ▢ **(c)** It sets the operating voltage of the processor.
- ▢ **(d)** It determines the font color of the output text.

Q47. What is the primary reason why 'Continuous Batching' achieves up to 4x higher system throughput than 'Static Batching' in multi-user LLM serving benchmarks?

- ▢ **(a)** It overclocks the server memory bus.
- ▢ **(b)** It eliminates bubbles and idle time by allowing completed requests to depart the batch immediately and admitting new incoming requests at every decode iteration, keeping accelerator compute and memory pipelines fully saturated.
- ▢ **(c)** It drops every fourth word generated by the model.
- ▢ **(d)** It runs the model in reverse order.

Q48. When converting a PyTorch model to OpenVINO IR using `ov.convert_model()`, how does setting `example_input` assist the conversion process?

- ▢ **(a)** It provides a training dataset to retrain the model weights.
- ▢ **(b)** It enables PyTorch's `torch.jit.trace` engine to trace real tensor execution paths, deduce tensor shapes, and resolve dynamic control flow operations into a static computational graph.
- ▢ **(c)** It prints the model's licensing agreement to the console.
- ▢ **(d)** It measures the power consumption of the graphics card.

Q49. What is the primary technical distinction between Habana Gaudi 2 and contemporary GPU architectures regarding host CPU dependency?

- ▢ **(a)** Gaudi 2 requires 16 host CPUs per accelerator card.
- ▢ **(b)** Gaudi 2 offloads the entire execution graph, DMA scheduling, and inter-accelerator collective communications (HCCL over on-die RoCE) directly to on-chip control processors, minimizing host CPU overhead and preventing host-side scheduling bottlenecks during multi-node scaling.
- ▢ **(c)** Gaudi 2 cannot communicate with host CPUs at all.
- ▢ **(d)** GPUs do not require host CPUs, while Gaudi requires 100 host CPUs.

Q50. In an enterprise HPC cluster deployment, what is the ultimate goal of combining OpenVINO INT4 quantization on 5th Gen Intel Xeon Scalable CPUs with Habana Gaudi accelerators for distributed training?

- ▢ **(a)** To replace all cluster software with proprietary mainframe operating systems.
- ▢ **(b)** To establish an end-to-end cost-effective, high-throughput AI infrastructure lifecycle: training and fine-tuning large foundation models efficiently at scale on Habana Gaudi clusters over standard RoCE Ethernet, and serving high-volume quantized inference directly on ubiquitous Intel Xeon server nodes with AMX acceleration without requiring expensive dedicated GPUs for every inference instance.
- ▢ **(c)** To run only legacy Fortran weather simulations.
- ▢ **(d)** To eliminate the need for network switches and data storage.